

Document Of Understanding

100 Black Men of Orange County Website Application

Fully Functional Mobile App

Project Overview:

The proposed project is a comprehensive web-based platform designed to digitally manage all aspects of an educational mentorship program. This system will enable a seamless experience for students, mentors, sponsors, and administrative staff. Through this application, the organization will be able to manage student records, mentor interactions, attendance, scheduling, communication, sponsor engagement, and documentation, all within a secure and scalable digital environment.

The platform aims to enhance operational efficiency, transparency, and communication by integrating digital tools that replace manual and paper-based workflows. It ensures better coordination between stakeholders while promoting student development and sponsor visibility.

Project Objective:

- Deliver a scalable, secure, and intuitive platform to support mentorship and academic growth.
- Empower mentors with real-time access to student data.
- Streamline communication and attendance tracking.
- Create space for sponsors and highlight contributions.

Business Objectives

- Enable digital onboarding and management of students, mentors, and parents.
- Provide a secure portal for parent-student-mentor communication.
- Facilitate digital handling of documents, forms, attendance, and permissions.
- Increase engagement and transparency through real-time updates and private content sharing.

User Experience Objectives

- Multi-role login portals with customized dashboards.
- Interactive calendar with detailed event pages.
- Seamless communication between users via internal messaging.
- Simple and secure registration, enrollment, and document submission flows.
- Mobile-friendly, accessible, and responsive interface.

Operational Objectives

- Simplify enrollment through digital forms and electronic signatures.
- Maintain detailed student profiles accessible by authorized mentors and parents.
- Manage attendance and provide absence reporting from the parent portal.
- Automate notifications via SMS and email for reminders.

Scope Definition Summary

In Scope:

- Multi-role login portal
- Messaging between students, mentors, and super admin
- Student profile management and progress tracking
- File upload system for report cards, community hours, etc.
- Digital consent forms
- E-signature functionality
- Attendance module with absence reporting
- Sponsor public listing and sign-up form
- Public blog/newsletter and private gallery
- Real-time SMS/email notifications
- Financial transactions/payment modules

User Types and Roles:

1. Student Flow

- **Step 1: Login**
Students authenticate using their unique student ID and password. The login system checks credentials and grants access to the student dashboard.
- **Step 2: Dashboard Access**
After successful login, students land on their personalized dashboard which includes attendance stats, recent notes from mentors, upcoming events, and document submission options.
- **Step 3: Profile Interaction**
Students can view their profile information, edit limited fields (like contact details), and upload documents such as community service hours or academic reports.
- **Step 4: Messaging System**
Students use a secure internal messaging feature to communicate directly with their assigned mentors or reach out to the admin/coordinator for queries.
- **Step 5: Event Calendar**
From their dashboard, students access a calendar that shows daily events. Clicking on a date reveals full event details such as venue, session topics, and lunch menus.

2. Mentor Flow

- **Step 1: Login**
Mentors log in through their designated portal. Their access is validated and directed to a mentor-specific dashboard.
- **Step 2: Dashboard Overview**
The dashboard displays key stats: list of assigned students, unread messages, upcoming mentoring sessions, and alerts from the admin.
- **Step 3: Student Profile Access**
Mentors click into individual student profiles to view uploaded documents, academic progress, attendance logs, and previously entered notes.
- **Step 4: Add Notes**
Using a secured input field, mentors record progress notes, behavioral observations, or action items. These are time-stamped and visible to both the student and super admin.
- **Step 5: Communication**
Mentors can message assigned students directly, or escalate communication to the program coordinator or administrator for further action.
- **Step 6: Calendar View**
The private calendar displays upcoming mentoring sessions, special events, or any internal meetings relevant to the mentor.

3. Super Admin Flow

- **Secure Login & Role Access**
MFA-enabled login with full control over platform configurations, access roles, and permissions.
- **User & Role Management**
Create, update, or deactivate any user type. Assign or revoke permissions.
- **Calendar & Content Control**
Manage system-wide calendars and media content. Control blog/news visibility and publication.
- **Compliance & Notifications**
Monitor user actions, verify uploaded documents, dispatch alerts, and access full audit logs.
- **Backup & Monitoring**
Trigger backups, restore data, and monitor system health and activity.

4. Sponsor Flow

- **Step 1: Access Public Page**
Sponsors visit the publicly accessible sponsorship page.
- **Step 2: Apply as a Sponsor**
Sponsors complete an online form including name, organization, contact info, and optional media (logo, description).
- **Step 3: Review & Approval**
Admins review sponsor submissions and approve or reject them.

- **Step 4: Visibility**

Approved sponsors are listed on a dedicated webpage that may include their logo, name, contribution type, and optional social links.

Platform

A web-based portal hosted in the cloud, supporting all roles with responsive UI.

Technology Stack

Frontend:

- React.js (UI Library)
- Tailwind CSS (Responsive and utility-first CSS framework)

Backend:

- Node.js (JavaScript runtime)
- Express.js or NextJS (Web server framework for API handling)

Database:

- PostgreSQL (Relational Database)

Authentication:

- JWT (JSON Web Token) for user session handling
- Role-based access control
- Optional multi-factor authentication for administrators

File Storage:

- AWS S3 or Firebase Cloud Storage (secure cloud storage of user-uploaded documents)

Email Service:

- SendGrid or AWS SES (for transactional emails and alerts)

SMS Service:

- Twilio (for sending real-time SMS notifications)

Hosting & Infrastructure:

- AWS EC2, GCP Compute Engine, or other preferred cloud platform
- Docker containers for service isolation and deployment
- CI/CD pipelines for code integration and automated deployments

Integrations:

- **Email:** SendGrid or AWS SES for transactional and broadcast emails
- **SMS:** Twilio for real-time text messaging and alerts
- **File Storage:** AWS S3 or Firebase for secure file upload and access
- **Authentication:** JWT-based system with optional MFA for admin roles
- **Hosting:** AWS EC2, GCP, or other cloud provider based on client preference

Users Web App Features:

1. Dashboard Interface

Each user role has a personalized dashboard showing relevant insights such as upcoming events, student updates, pending tasks, unread messages, and alerts.

2. User Authentication & Role-Based Access

Secure login for students, mentors, super admins, and sponsors, with access control that limits visibility and editing rights based on role.

3. Profile Management

Users can view and update their profiles. Students and mentors have editable fields and document upload capabilities, while admins can manage all user profiles.

4. Event Calendar

Users can view a calendar populated with program events. Clicking a date reveals detailed info: topic, time, location, session description, and downloadable resources.

5. Internal Messaging System

Allows direct communication between users within the app (e.g., student ↔ mentor, mentor ↔ super admin). Threads are stored and searchable.

6. Notifications Center

Users receive real-time alerts via in-app notifications, SMS, or email. Alerts include event reminders, new messages, file submission deadlines, and announcements.

7. Document Upload

Students and mentors can upload relevant documents (e.g., attendance slips, community service hours, report cards), which are stored securely and available for super admin review.

8. Digital Forms and Signatures

Online submission of registration, agreement, and permission forms with integrated electronic signature support.

9. Attendance Tracking

Students can view their attendance records. Admins and mentors can mark attendance or review absence reports submitted by users.

10. Sponsor Engagement Interface

A public form for sponsors to submit interest. Super admin-reviewed entries will appear on a public sponsor listing page, complete with profile and media.

Core Functional Modules

1. Authentication and Role-Based Access

- Secure user login system supporting different roles (super admin, mentor, student, sponsor).
- Role-based dashboards and access permissions.
- Password reset, email verification, optional MFA for admins.

2. Calendar with Embedded Event Info

- Public calendar for general announcements.
- Private calendars per user role.
- Event details embedded per date: agenda, location, materials, meal plans, etc.

3. Student Profiles and Progress Tracker

- Detailed student dashboard with academic progress, attendance, uploaded documents.
- Mentor notes attached with timestamps.

4. Digital Forms with E-signature

- Digital enrollment forms, agreements, and field trip permissions.
- Embedded e-signature and form validation.

5. Attendance Logging & Absence Reports

- Daily attendance dashboard.
- Manual absence reporting by student or super admin.

6. Messaging System (Internal)

- One-to-one and one-to-many communication.
- Mentors <-> Students, Super Admin <-> All.
- Message archiving, tagging, and alerts.

7. Document Upload and Viewer

- File upload interface for report cards, documents.
- Preview and download permissions based on role.

8. Newsletter and Blog Module

- Public-facing news page for events, sponsors, highlights.
- Admin-curated content with images and rich formatting.

9. Sponsor Signup and Management

- Sponsor onboarding form.
- Admin review workflow.
- Public listing with profile highlights.

10. Notification Engine (SMS/Email)

- Role-specific and event-based alerts.
- Integration with Twilio and SendGrid.

Security & Privacy:

- Encrypted password storage and secure session handling
- Role-based access control (RBAC) for all users
- HTTPS enforced throughout the application
- Secure file uploads with virus scanning
- GDPR-compliant data handling (where applicable)
- Logging and audit trails for sensitive actions
- Regular backups and disaster recovery mechanisms

Development Phases & Timeline

Phase	Duration
Requirement Gathering	1 Week
UI/UX Design	6 Weeks
Development	12 Weeks
Testing & QA	2 Weeks
Deployment	4 Week
Post-launch Support	60 Days

Deliverable:

- Functional, role-based web portal
- File management and secure cloud storage
- Dashboard for each user type
- Notification and messaging engines
- Final deployment with user training
- 2 months post-launch support and maintenance (included)

Total Project Cost Estimate

The total estimated project cost is **\$5,000**, broken down into the following key phases:

Phase	Description	Payment %	Amount (USD)
Upfront Booking	Project initiation, team allocation, and planning	50%	\$2,500
Design & Architecture	Wireframes, UI design, architecture documentation	25%	\$1,250
Core Development	Backend, frontend, and database implementation	25%	\$1,250
Total Project cost		100%	\$5,000

Web Application Maintenance Plans

To ensure continued performance, security, and scalability of the web application after launch, we offer tiered maintenance package tailored to different operational needs and budgets. This package is designed to maintain the integrity of the system while offering increasingly robust service levels, from essential support to high-security enterprise-grade protection.

1. Gold Plan – \$400/month

Built for institutions prioritizing high-security, compliance, and expert cyber defense. This tier includes access to professional ethical hackers and white-hat vulnerability experts.

Included Services (Gold +):

- Quarterly penetration testing by certified ethical hackers (CEH/CPT)
- Advanced threat detection systems (e.g., WAF, IDS/IPS setup)
- 24/7 monitoring for intrusion attempts and threat intelligence feeds
- API rate-limiting & brute-force attack prevention strategies
- Zero-day exploit advisory alerts with patch strategies
- Executive report with security health metrics
- Dedicated account engineer

Hosting & Server Options

Option 1: Amazon Web Services (AWS)

- **Compute:** EC2 instances (scalable on-demand virtual servers).
- **Storage:** S3 for files, RDS for relational database.
- **Strengths:** Market leader, wide ecosystem, highly reliable, proven at scale.
- **Pricing:** Pay-as-you-go, granular cost control, free tier available. Typically \$12 – \$15 per month for small to medium-scale deployments. (Pricing may vary as per the package you opt for.)

Best For: Organizations prioritizing global infrastructure, integrations with third-party tools, and mature DevOps pipelines.

Option 2: Microsoft Azure

- **Compute:** Azure Virtual Machines, App Service for web hosting.
- **Storage:** Azure Blob Storage, Azure SQL Database.
- **Strengths:** Seamless integration with Microsoft 365, Active Directory, and enterprise environments. Strong compliance certifications.
- **Pricing:** Competitive, with enterprise licensing discounts if already using Microsoft ecosystem. Typically \$20 – \$25 per month for small to medium-scale deployments. (Pricing may vary as per the package you opt for.)

Best For: Institutions with existing Microsoft products and enterprise IT setups.

Recommendation:

If long-term scale and integrations matter → AWS

If you rely on Microsoft tools (Office 365, Teams, Outlook, Active Directory) → Azure Both support CI/CD, global deployment, security compliance (HIPAA, GDPR), and disaster recovery.

EIN & ITIN Numbers (Company Information)

Company Information

- EIN (Employer Identification Number): **38-4277690**
- ITIN (Individual Taxpayer Identification Number): **901-50-3944**

This ensures compliance for tax, invoicing, and client documentation.